



Raspodijeljene i nerelacijske baze podataka

VJEŽBA 05: JSON

OFFSET *x* ROWS → preskače *x* broj redaka

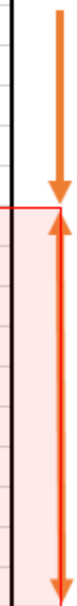
FETCH NEXT *y* ROWS ONLY → dohvaća sljedećih *y* redaka

Parcijalni dohvat podataka

- Napravite upit kojim dohvaćate podatke iz pogleda `production.v_products_stock`
- Kako bi dohvatili prvih 10 redaka?
- A sljedećih 10?
- ...sljedećih 10...

```
SELECT *  
FROM production.v_products_stock  
ORDER BY 1  
      OFFSET 5 ROWS  
      FETCH NEXT 10 ROWS ONLY;
```

ID	Name
1	Item #1
2	Item #2
3	Item #3
4	Item #4
5	Item #5
6	Item #6
7	Item #7
8	Item #8
9	Item #9
10	Item #10
11	Item #11
12	Item #12
13	Item #13
14	Item #14
15	Item #15
16	Item #16
17	Item #17
18	Item #18
19	Item #19
20	Item #20



OFFSET 5 ROWS

FETCH NEXT 10 ROWS ONLY

Zadatak

- Napravite proceduru `sp_products_paging` koja prikazuje naziv proizvoda i količinu na skladištu a kao parametre prima:
 - id trgovine (`store_id`),
 - broj redaka koji se preskače (OFFSET)
 - broj redaka koji se želi dohvatiti.

```
CREATE PROCEDURE [dbo].[sp_products_paging]
(@store_id INT ,@skip INT, @getRows INT)
AS
BEGIN

SELECT product_name, quantity
FROM production.v_products_stock
WHERE store_id= @store_id
ORDER BY 1
OFFSET @skip ROWS
FETCH NEXT @getRows ROWS ONLY;

END
```

Zadatak

- Preuredite proceduru `sp_products_paging` tako da ako parametri nisu uneseni predefinirane (default) vrijednosti za preskakanje i dohvaćanje budu 10 i 10:

```
CREATE PROCEDURE [dbo].[sp_products_paging]
(@store_id INT ,@skip INT =10, @getRows INT =10)
AS
BEGIN

SELECT product_name, quantity
FROM production.v_products_stock
WHERE store_id= @store_id
ORDER BY 1
OFFSET @skip ROWS
FETCH NEXT @getRows ROWS ONLY;

END
```

Zadatak

- Napravite upit koji prikazuje ukupnu količinu svakog proizvoda na skladištu

```
SELECT
p.product_name,
sum(s.quantity) total
from production.stocks s
inner join
production.products p
on
s.product_id=p.product_id
GROUP BY product_name
```

	product_name	total
1	Electra Amsterdam Fashion 3i Ladies' - 2017/2018	47
2	Electra Amsterdam Fashion 7i Ladies' - 2017	62
3	Electra Amsterdam Original 3i - 2015/2017	11
4	Electra Amsterdam Original 3i Ladies' - 2017	12
5	Electra Amsterdam Royal 8i - 2017/2018	45
6	Electra Amsterdam Royal 8i Ladies - 2018	26
7	Electra Cruiser 1 - 2016/2017/2018	34
8	Electra Cruiser 1 (24-Inch) - 2016	81
9	Electra Cruiser 1 Ladies' - 2018	10
--	-- -- -- -- --	--

```
SELECT product_name,
sum(quantity) total
FROM
production.v_products_stock
GROUP BY product_name
```

Proučite pa pokrenite proceduru: sp_get_json_products

Zadatak

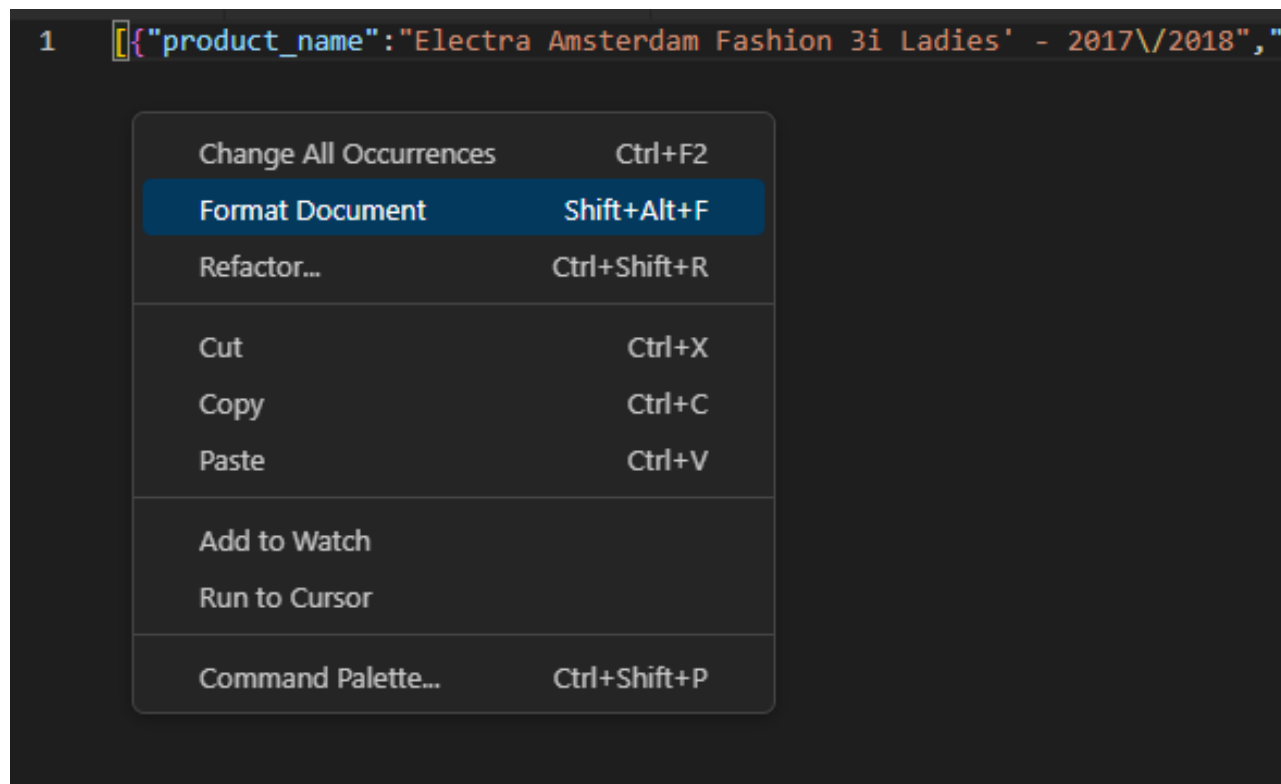
- Proučite pa pokrenite proceduru: `sp_get_json_products`
- Proučite sljedeći SELECT i provjerite što vraća

```
SELECT product_name, SUM(quantity) total  
FROM production.v_products_stock  
GROUP BY product_name  
FOR JSON PATH
```

Rezultate zalijepite u VS Code
usporedite ih .

```
DECLARE @result VARCHAR(MAX);  
  
SET @result = (SELECT product_name, SUM(quantity) total  
FROM production.v_products_stock GROUP BY product_name  
FOR JSON AUTO)  
  
SELECT @result;
```

- Odaberite jezik JSON



```
{
  "product_name": "Electra Amsterdam Fashion 3i Ladies' - 2017\\2018",
  "total": 47
},
{
  "product_name": "Electra Amsterdam Fashion 7i Ladies' - 2017",
  "total": 62
},
{
  "product_name": "Electra Amsterdam Original 3i - 2015\\2017",
  "total": 11
},
{
  "product_name": "Electra Amsterdam Original 3i Ladies' - 2017",
  "total": 12
},
{
  "product_name": "Electra Amsterdam Royal 8i - 2017\\2018",
  "total": 45
},
{
  "product_name": "Electra Amsterdam Royal 8i Ladies - 2018",
  "total": 26
},
}
```

Prikaz NULL vrijednosti

- Napravite upit koji vraća popis **svih** proizvoda i ukupan broj istih na skladištu. Generirajte JSON iz istog upita i uočite kako se prikazuju NULL vrijednosti.

```
SELECT
p.product_name,
SUM(s.quantity) total
FROM production.stocks s RIGHT JOIN
production.products p
ON s.product_id=p.product_id
GROUP BY product_name
ORDER BY 2
FOR JSON AUTO , INCLUDE_NULL_VALUES
```

```
{
  "product_name": "Trek Checkpoint SL 6 - 2019",
},
{
  "product_name": "Trek1",
},
{
  "product_name": "Trek Domane SLR Frameset - 2018",
  "total": 5
},
{
  "product_name": "Electra Superbolt 1 20\" - 2018",
  "total": 9
}
```


Dodatne možnosti

```
SELECT  
p.product_name,  
SUM(s.quantity) total  
FROM production.stocks s RIGHT JOIN  
production.products p  
ON s.product_id=p.product_id  
GROUP BY product_name  
ORDER BY 2  
FOR JSON AUTO
```

```
, INCLUDE_NULL_VALUES  
, WITHOUT_ARRAY_WRAPPER  
, ROOT('Products')
```

Dodatna kontrola

- Napravite upit koji vraća naziv trgovine, naziv proizvoda i količinu proizvoda na skladištu trgovine pa generirajte JSON.

```
SELECT
st.store_name,
p.product_name,
s.quantity
FROM production.stocks s INNER JOIN
production.products p
ON s.product_id=p.product_id
INNER JOIN sales.stores st ON
st.store_id=s.store_id
FOR JSON AUTO;
```

```
[
  {
    "store_name": "Santa Cruz Bikes",
    "p": [
      {
        "product_name": "Trek 820 - 2016",
        "s": [
          {
            "quantity": 27
          }
        ]
      },
      {
        "product_name": "Ritchey Timberwolf Frameset - 2016",
        "s": [
          {
            "quantity": 5
          }
        ]
      }
    ]
  }
]
```

Dodatna kontrola

- Usporedite prikaz

FOR JSON AUTO;

```
[
  {
    "store_name": "Santa Cruz Bikes",
    "product_name": "Trek 820 - 2016",
    "quantity": 27
  },
  {
    "store_name": "Santa Cruz Bikes",
    "product_name": "Ritchey Timberwolf Frameset - 2016",
    "quantity": 5
  },
]
```

FOR JSON PATH;

```
[
  {
    "store_name": "Santa Cruz Bikes",
    "p": [
      {
        "product_name": "Trek 820 - 2016",
        "s": [
          {
            "quantity": 27
          }
        ]
      },
      {
        "product_name": "Ritchey Timberwolf Frameset - 2016",
        "s": [
          {
            "quantity": 5
          }
        ]
      }
    ]
  },
]
```

Dodatna kontrola

```
SELECT
st.store_name,
p.product_name AS 'product.name',
s.quantity AS 'product.quantity'
FROM production.stocks s INNER JOIN
production.products p
ON s.product_id=p.product_id
INNER JOIN sales.stores st ON
st.store_id=s.store_id
FOR JSON PATH;
```

```
[
  {
    "store_name": "Santa Cruz Bikes",
    "product": {
      "name": "Trek 820 - 2016",
      "quantity": 27
    }
  },
  {
    "store_name": "Santa Cruz Bikes",
    "product": {
      "name": "Ritchey Timberwolf Frameset - 2016",
      "quantity": 5
    }
  }
],
```

Dodatna kontrola

- Usporedite formate

```
[
  {
    "store_name": "Santa Cruz Bikes",
    "product": {
      "name": "Trek 820 - 2016",
      "stock": {
        "quantity": 27
      }
    }
  },
  {
    "store_name": "Santa Cruz Bikes",
    "product": {
      "name": "Ritchey Timberwolf Frameset - 2016",
      "stock": {
        "quantity": 5
      }
    }
  }
],
```

```
SELECT
st.store_name,
p.product_name AS 'product.name',
s.quantity AS 'product.stock.quantity'
FROM production.stocks s INNER JOIN
production.products p
ON s.product_id=p.product_id
INNER JOIN sales.stores st ON
st.store_id=s.store_id
FOR JSON PATH;
```

```
[
  {
    "store_name": "Santa Cruz Bikes",
    "product": {
      "name": "Trek 820 - 2016",
      "quantity": 27
    }
  },
  {
    "store_name": "Santa Cruz Bikes",
    "product": {
      "name": "Ritchey Timberwolf Frameset - 2016",
      "quantity": 5
    }
  }
],
```

OPENJSON

```
DECLARE @json NVARCHAR(MAX);  
SET @json = (SELECT category_name FROM production.categories FOR JSON PATH)  
  
SELECT * FROM OPENJSON(@json)  
WITH (  
    category_name    varchar(200) '$.category_name'  
) ;
```

Zadatak

- Napravite proceduru koja prima parametra tipa varchar `sp_ins_categoryJSON` u kojem će biti pohranjene kategorije u JSON formatu. U proceduri će se podaci iz ulaznog parametra pohranjivati u tablicu `production.categories`
- Primjer pozivanja procedure:

```
exec sp_ins_categoryJSON ' [{"category_name": "Trekking  
Bicycles"}, {"category_name": "Hybrid/Commuter Bikes"} ] '
```

Zadatak

```
CREATE PROCEDURE sp_ins_categoryJSON(@catJSON varchar(MAX))
AS
BEGIN

INSERT INTO production.categories
SELECT * FROM OPENJSON(@catJSON)
WITH (
    category_name    varchar(200) '$.category_name'
)

END
```

```
exec sp_ins_categoryJSON ' [{"category_name":"Trekking  
Bicycles"} , {"category_name":"Hybrid/Commuter Bikes"} ] '
```